

LEOPARD GECKO · EUBLEPHARIS
MACULARIUS

Keep your leopard gecko the right way from day one.

A complete, fact-checked printable manual: belly heat, the three-hide humidity system, optional UVB (ultraviolet B) lighting, feeding by age, handling and body language, health triage, the law in every state, and the routines that keep a leopard gecko thriving for its full lifespan.

INSIDE

- ✓ Belly heat, humidity and UVB targets in Fahrenheit and Celsius
- ✓ Feeder, gut-load and never-feed charts
- ✓ Stick tail, bone disease and impaction triage, a symptom table and an emergency card
- ✓ Owner logs, a pet-sitter sheet and a power-outage plan

48 pages

Beginner

Print or tablet

Edition 3.0 · Oct 2026



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Every page is written to be used, not just read once.

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START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

UNITS USED THROUGHOUT

Every temperature and size is given twice. Temperatures appear in degrees Fahrenheit (°F) with degrees Celsius (°C) beside them. Sizes appear in inches (in) with centimeters (cm) beside them, and tank sizes in gallons, the way they are sold. A gecko and its weight log are measured in grams (g), the unit a kitchen scale shows. Humidity is a percentage (%) of the moisture the air could hold. Prices are in United States dollars.

If you haven't set up yet

Check the law first, on page 25: a leopard gecko is legal without a permit in most of the United States, but not everywhere. Then read pages 4 to 11 in order before buying anything. The enclosure size and the three-hide layout decide most of what goes in the cart, and the thermostat is never optional. Use the Budget & Shopping List on page 35 to buy once.

If your gecko is already home

Audit what you have against the Setup Checklist in the print list below, and go to the Red Flags page (page 26) if anything seems off. Read the warm hide floor with an infrared thermometer before you change anything else: a warm side running too cool explains more refused meals than any other fixable cause, and the feeding pages starting on page 12 show why.

Print these first

- | | |
|--|--|
| ☐ Page 33: Setup Checklist & Targets | ☐ Page 40: Pet-Sitter Sheet |
| ☐ Page 34: Emergency & Quick Targets Card | ☐ Page 41: Owner Log |
| ☐ Page 37: Symptom Quick Reference | ☐ Page 42: Equipment & Vet Log |
| ☐ Page 38: Daily, Weekly & Seasonal Routine | ☐ Page 43: Enrichment Checklist & Log |

WHY THIS PACKAGE EXISTS

Leopard geckos are sold as the easy reptile, and the setup sold with one is often years out of date: a 20 gallon tank, calcium sand, one hide, no thermostat, no UVB. This package reflects current enclosure-size, heating, humidity and diet standards, cross-checked against the veterinary and scientific sources listed on page 46.

NOT A SUBSTITUTE FOR

A reptile-experienced veterinarian. This is a husbandry reference for day-to-day care. Anything urgent or health-related is always a vet call, not a page in this book. Find a reptile vet before you need one and write the number on the emergency card. Nothing on the law page is legal advice either.

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Eublepharis macularius</i>
Adult size	6 to 9 in (15 to 23 cm) nose to tail; males 60 to 90 g, females as light as 45 g
Captive lifespan	10 to 20 years with proper care, and some reach 25 to 30
Native range	Rocky desert, arid grassland and dry scrub in Pakistan, Afghanistan, Iran, India and Nepal
Diet type	Obligate insectivore: live insects only, no fruit or vegetables (page 12)
Active	At dusk and dawn (crepuscular), so evening is feeding and handling time
Enclosure minimum (adult)	36×18×18 in (91×46×46 cm), front-opening preferred; floor space matters more than height
Heat	An under-tank heat mat on a thermostat, 88 to 92°F (31 to 33°C) on the warm hide floor
UVB	Optional but beneficial, kept low: a UVI (ultraviolet index, the strength of the UVB) of 0.6 to 1.4 at the basking spot
Handling difficulty	Beginner; tolerant once settled, and never picked up by the tail
Conservation status	Least Concern, the lowest-risk grade on the global list of threatened species

Cost overview

Initial setup

\$20 to \$45 for a pet-store gecko, \$250 to \$400 for the equipment in most first-time setups, plus \$150 to \$215 for the first vet exam and fecal test (a stool check for parasites). Every line is on page 35.

Monthly ongoing

\$20 to \$50 for most owners: feeder insects, supplements, substrate, electricity and UVB tube replacement. The yearly wellness exam, \$85 to \$105, comes on top.

BUDGET FOR THE WORST CASE TOO

A sick or emergency visit runs \$100 to \$800 or more, and diagnostics or surgery push it higher again. Treat a small emergency fund as part of the setup cost, not an optional extra.

Fun facts worth knowing

- Unlike most geckos, leopard geckos have movable eyelids and can blink. They also lack the sticky toe pads other geckos climb glass with.
- The fat tail is the gecko's energy store. A plump tail means good condition, and a thinning one is often the first sign that something is wrong.
- The temperature an egg is incubated at decides the sex: cooler eggs hatch mostly females and warmer eggs mostly males.

The species, morphs, size & lifespan

What you are buying, how big it gets, and how long it stays.

One species

The pet leopard gecko is *Eublepharis macularius*. In the wild it lives across arid country from Pakistan to northwestern India, sheltering by day in rocky outcrops and dry grassland and coming out at dusk to hunt insects, scorpions, small lizards and worms. It is a ground dweller: no toe pads, no climbing glass, and a life spent on and under the floor, which is why floor space matters more than height in its enclosure. It is not a social species and does not do well sharing an enclosure, so every leopard gecko is housed alone.

Morphs

A morph is a captive-bred line that looks different from the natural, wild-type gecko: a different color or pattern. Over 100 are recognized. A standard or wild-type gecko costs \$20 to \$45 at a pet store; breeders often charge \$50 to \$150 even for a basic morph, popular morphs sit around \$75 to \$300, and rare designer morphs run \$500 to \$3,500 or more. The care in this package applies to every morph with one exception. Albino and other pale morphs burn more easily under UVB, so they get a UVI of 0.5 to 0.7 or lower instead of 0.6 to 1.4 (page 9).

Size

Adult length	6 to 9 in (15 to 23 cm) nose to tail tip.
Adult weight	Males 60 to 90 g. Females are lighter, some as light as 45 g. A gecko that drops its tail loses a large share of its reserves at once, which is why the tail matters so much in an animal this small.
Young geckos	A gecko under about 3 in (8 cm) long is fragile and easily injured. It waits until it reaches roughly 6 in (15 cm) before regular handling.
Counted as adult	Once the tail has thickened to about the width of the neck. That is the point it moves onto the adult feeding schedule (page 12).

Lifespan

10 to 20 years is typical in captivity with proper care, and some leopard geckos reach 25 to 30. Wild geckos manage 6 to 8 years, because predation, parasites and harsh conditions cut them short. At \$250 to \$700 a year in running costs, 10 to 20 years adds up to \$2,500 to \$14,000 before a single sick visit. A hatchling bought today may outlast its enclosure, its heat mat and several UVB tubes, so plan for the whole span: the routine and a reptile vet for every one of those years.

CAPTIVE-BRED IS THE ONLY KIND ON SALE

Leopard geckos were imported into the United States and Europe from Pakistan, Afghanistan and India through the 1960s and 1970s, and were being bred in the United States by the early 1970s.

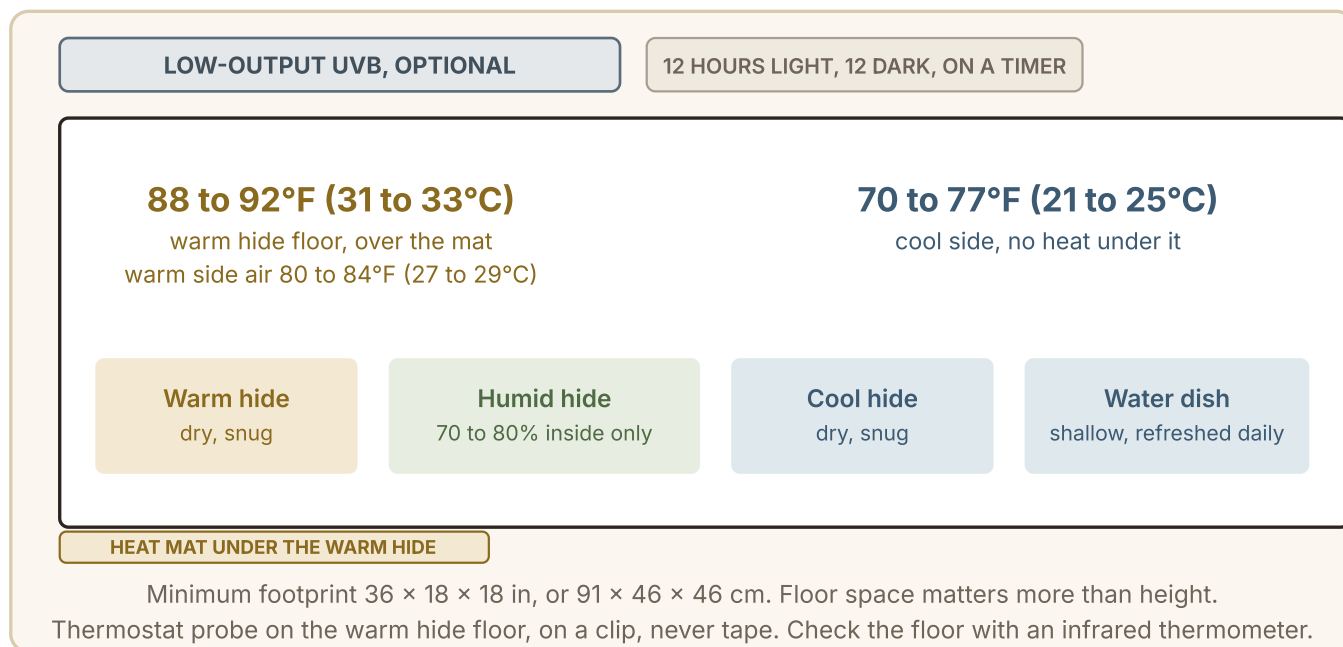
Large-scale breeding started in the late 1980s, and the color morphs followed in the 1990s.

Essentially every leopard gecko sold today is captive-bred, with no wild-collected supply behind the pet trade. What to look for in the animal itself is on page 21.

Enclosure size, type & where it goes

36×18×18 in (91×46×46 cm) is the current minimum for one adult, and bigger is always fine. The 20 gallon long tank, roughly 30×12×12 in (76×30×30 cm), is the old floor and no longer the standard. A hatchling can start in something as small as a 10 gallon while it is small, then move up. Leopard geckos are terrestrial and cannot climb smooth glass, so floor space matters far more than height.

Choose a front-opening, well-ventilated enclosure with an escape-proof screen top. A PVC (polyvinyl chloride, a rigid plastic) or wood enclosure keeps heat that a glass tank loses, so a heat setting carried over from glass lands hotter: read the warm hide floor again after any change of enclosure (page 8).



NEVER

Run a heat mat without a thermostat. An unregulated mat is a real burn and fire risk, and unregulated heat is one of the most common causes of thermal burns in captive leopard geckos. Skip hot rocks entirely: they heat unevenly and burn.

Where it goes

- Out of the kitchen and away from anywhere food is made or served.
- In a room that stays at 65°F (18°C) or warmer at night if you can, so no night heat is needed.
- One gecko per enclosure. This is not a social species, and it does not do well sharing.

What the diagram doesn't show

- Hides should be snug and cave-like, never oversized and open. A roomy hide gives no sense of cover.
- Add rock or wood for nail filing and light climbing.
- The humid hide sits on the cool or middle side, never over the mat (page 10).

The temperature gradient & belly heat

A warm end, a cool end, and a gecko that moves between them to set its own body temperature.

ZONE	TARGET (°F)	TARGET (°C)
Warm hide floor (belly heat)	88 to 92	31 to 33
Warm side, ambient air	80 to 84	27 to 29
Cool side	70 to 77	21 to 25
Night, whole enclosure	the 70s; 65 is the floor	21 to 26; 18 is the floor
Humidity	30 to 40% through the enclosure, 70 to 80% inside the humid hide only	

The warm hide floor is the number to build everything else around. A floor surface runs above the air around it, which is why the floor target sits above the ambient one.

Why the floor number matters

A leopard gecko cannot make its own heat. Everything it does, from digesting a cricket to fighting off an infection, runs on warmth borrowed from the enclosure floor, and it digests on its belly. A gecko kept a few degrees too cool goes off food long before it looks sick. It moves between the warm and cool zones through the day to manage digestion and activity, so a tank warm from end to end takes that choice away and stresses it over time.

Measuring

- Read the warm hide floor with an infrared thermometer, a handheld gun aimed at the surface the gecko lies on. A thermostat set to 90 does not mean the floor is 90; the infrared reading is how you find out.
- Use digital thermometers with probes for the air in each zone, and a hygrometer (humidity gauge). A stick-on dial reads the air at the glass, not the floor.
- Check at different times of day. Leopard geckos are active at dusk and dawn, so night temperatures matter too.

Signs the temperatures are wrong

YOU NOTICE	LIKELY MEANS
Off food and moving less, but looks well and the tail is full	Too cold. It shows slowly, as appetite and movement, before anything you can see on the animal
Open-mouth breathing, nasal discharge, labored breathing in a cool enclosure	Possible respiratory infection: a vet visit, not a heating tweak (page 30)
Glass surfing or pacing away from the warm side, gaping or panting, refusing the warm hide, flattening into the coolest spot, sluggish or oddly restless	Too hot, with heat it cannot escape
Darkened skin, wobbliness or limpness	An emergency: move it somewhere cooler, switch the heater off, and call a reptile vet

Either way, the first step is the same: read the floor with the infrared thermometer, read the cool side with a probe, and fix the number before you change anything else.

Heat mats, thermostats, probes & timers

Belly heat first

An under-tank heat mat under the warm hide, on a thermostat, is the standard here. A mat makes a small, steady warm patch, uses little power, and can run all night without light. Overhead heat, a halogen bulb or a deep heat projector (a lightless heater that sends its heat down in a focused beam), warms a wider area but dries the air and can burn a gecko that climbs too close. Many setups run both: overhead by day for the gradient, the mat for the floor. Belly heat is preferred; skip an overhead lamp as the only source.

The thermostat is the safety device

A thermostat measures the temperature at one point and cuts or lowers the power once that point reaches the setting. Without one, a mat or bulb draws full power with no ceiling, and a gecko pressed against a hot surface cannot always move away. Every heater gets one.

HEAT SOURCE	CONTROLLER
Under-tank heat mat	On/off thermostat with a probe. A mat changes temperature slowly enough that a full on and off cycle doesn't make a harsh swing.
Halogen bulb or deep heat projector	Dimming (proportional) thermostat. It eases the power down before the surface overshoots.
Ceramic heat emitter (night heat)	Pulse proportional or dimming thermostat. It reacts faster than a mat, so a plain on/off unit can overshoot before it cuts.

Where the probe goes

On the floor inside the warm hide, directly over the mat, on the surface the gecko lies on. Hold it with a probe clip or a suction cup, never tape: tape over a probe throws off its reading, and a gecko can stick to exposed adhesive and tear its skin. A probe stuck to the glass, hanging in the air, or sitting on the cool side holds the wrong number all year.

How heating fails

- **The probe moves.** The most common failure, and a silent one: the display keeps showing an accurate reading for the wrong spot. Check the probe's position at every spot clean.
- **The mat makes a hot spot away from the probe.** A thermostat will not catch it. Scan the whole warm end with the infrared thermometer rather than trusting the display.
- **The thermostat sticks on.** Rarer and more dangerous. Keep a digital thermometer at the warm end, apart from the probe, and read it daily: a reading that disagrees sharply with the display, or keeps climbing, is the warning.
- **Check before the gecko goes in,** with both the probe thermometer and the infrared gun, and again after any change of enclosure, substrate or hide.

Clearance, timers and night heat

Any overhead heat source or lamp sits at least 4 to 6 in (10 to 15 cm) above the highest point the gecko can reach, behind a guard. Put the lights on a plug-in timer for 12 hours on and 12 off, so day length never depends on you remembering. Skip colored night bulbs, red and blue included: they disrupt sleep. A light bulb cannot be night heat. If the room drops below 65°F (18°C), use a ceramic heat emitter, which gives heat with no light, on its own thermostat; a room that holds 65°F needs no night heat at all.

UVB: low output, optional & worth it

Not required the way it is for a bearded dragon. A real benefit anyway.

Why bother

Leopard geckos were kept without UVB for decades on the assumption that a crepuscular species gets all the vitamin D3 it needs from its food. Controlled work has since found otherwise: Gould and colleagues (2018) and Oonincx and colleagues found that even brief, low-level UVB, as little as two hours a day, measurably raises vitamin D3 made in a leopard gecko's skin. Vitamin D3 is what lets the body absorb calcium. This species appears to rely heavily on UVB and correct temperature to process calcium, not on supplements alone, which is part of why metabolic bone disease turns up in geckos kept without it.

UVI targets

Leopard geckos are Ferguson Zone 1, the crepuscular and shade-dwelling group. Aim for a UVI of 0 to 0.7 across most of the enclosure, peaking at **0.6 to 1.4** at the basking spot, or **0.5 to 0.7 or lower** for albino and other pale morphs, which burn more easily. Deep shade has to be available, and the hides provide it.

Which light

A linear T5 HO tube (a high-output fluorescent tube 5/8 inch across) from the low-output end of the range, mounted at a generous distance. A tube lays a gradient across the floor. Compact coil bulbs give a less even spread and have been linked to eye and skin irritation.

Distance, mesh and glass

Distance is part of the dose: UVB falls off sharply as the tube goes higher, so the same tube is strong close up and weak far away. Set the height from the tube's own distance chart for the UVI above. Glass or plastic between tube and gecko blocks most UVB, and a screen top cuts output substantially, so a tube above mesh needs that mesh accounted for. Like any lamp, keep it at least 4 to 6 in (10 to 15 cm) above the highest point the gecko can reach.

MEASURE, DON'T ASSUME

A handheld UVI meter reads what actually reaches the basking spot rather than what the box claims. It is the most expensive item in a reptile kit and genuinely optional, but it is the only way to know rather than assume.

Schedule and replacement

- On the same 12-hour timer as the rest of the lighting.
- Replace every 6 to 12 months depending on the model, by the calendar and the maker's rated life. Output fades long before the light dims, so a year-old tube that still lights may be giving almost no UVB. Write the date on page 42.
- No UVB at all is a choice some keepers make. If you make it, the calcium with vitamin D3 dust becomes the backup the gecko depends on (page 16).

Substrate, furnishings & the three hides

Substrate

Paper towel or tile are the safest choices, and the right default for anyone newer to reptiles: they remove the impaction risk and make a daily stool check easy. More experienced keepers sometimes move to a bioactive sand or soil mix, a planted floor with tiny invertebrates that break down waste, but that has more to get right and suits healthy adults only. A dig box (page 20) gives a gecko on paper or tile somewhere to dig.

NEVER

Calcium sand, or any fine loose sand. It carries a real impaction risk (page 29), whatever the label says.

The three hides are the husbandry, not the decoration

A warm dry hide over the mat, a cool dry hide at the far end, and a humid hide on the cool or middle side are the minimum. Together they let a gecko move between temperatures and moisture levels without ever leaving cover, which is a real choice rather than a forced trade between feeling safe and being the right temperature. Keep all three snug and cave-like.

The humid hide

A closed hide lined with damp substrate, running 70 to 80% humidity inside while the rest of the enclosure stays at 30 to 40%. It prevents the majority of shedding problems in this species. Keep the lining damp, not swampy and not bone dry, and replace it before it breaks down. Don't line it with sphagnum moss or paper towel the gecko could swallow. Don't raise the humidity of the whole enclosure to reach 70% either: that just adds a respiratory infection risk.

Furnishings checklist

- | | |
|--|--|
| ☐ Warm dry hide directly over the mat, probe inside it | ☐ Rock, slate or wood for nail filing and light climbing |
| ☐ Cool dry hide at the far end | ☐ Cork flats or low sturdy branches to rest off the floor |
| ☐ Humid hide with damp lining, cool or middle side | ☐ A small dish of plain calcium, an optional backup (page 16) |
| ☐ Shallow water dish the gecko can climb in and out of, refreshed daily | ☐ Digital thermometer and hygrometer |

HUMIDITY, IN ONE LINE

Put the moisture in one box, not in the air. Glass fogging or a damp smell means the enclosure itself is too humid: fix the ventilation and look for the source, such as a humid hide that has soaked the substrate around it.

Cleaning & hygiene

Cleaning removes the waste. Disinfecting kills what is left. Hygiene keeps it off your hands.

SALMONELLA

Healthy reptiles shed *Salmonella* bacteria in their droppings without being ill, on and off, and one negative test clears nothing. It travels from the gecko, its enclosure and anything it walks on to hands, then to mouths or food, so wash with soap and running water after every contact. Do not kiss the gecko or hold it to your face. Children under 5 should not handle the gecko or touch anything in its enclosure, and adults over 65, pregnant people and anyone with a weakened immune system are at higher risk. Geckos are not exempt: an outbreak linked to pet geckos, closed in January 2026, made 113 people sick across 36 states and put 31 in the hospital.

Three clocks

WHEN	WHAT TO DO
Daily	Remove droppings, urates (the white part of the waste), uneaten insects and shed skin as soon as you see them. Empty, wash and refill the water dish with fresh dechlorinated water.
Weekly	Wipe down glass, walls, hides and decor with a reptile-safe cleaner.
Monthly	Move the gecko to a secure temporary tub. Take everything out, scrub it with detergent and hot water, disinfect the enclosure, hides and dishes, rinse, dry, and put in fresh substrate.

Paper towel is replaced whenever it is soiled. A bioactive enclosure skips the monthly strip-down: pick up what you can see and let the cleanup crew handle the rest.

Disinfecting, step by step

- **Scrub first.** A film of waste shields bacteria from disinfectant, so it only works on a surface already cleaned.
- **Dilute household bleach 1:32**, which is half a cup of bleach per gallon of water. It is inexpensive and works against bacteria and many viruses.
- **Keep it wet for 10 minutes.** Bleach needs that long in contact to work, so don't wipe it straight off.
- **Rinse until no bleach smell remains**, then let the enclosure and every item dry completely and air out before the gecko goes back in. Plan on a full day of airing.

NEVER THE KITCHEN SINK

Never wash the gecko, its dishes, hides or enclosure in a kitchen sink, a bathroom sink, or a bath people use, or anywhere food is prepared. Keep a plastic tub for the job, or work in a utility sink or outdoors. If a bathtub does get used, clean it thoroughly afterward. Wash hides and decor separately from anything used for food, and keep the gecko out of the kitchen entirely.

If someone in the house gets sick

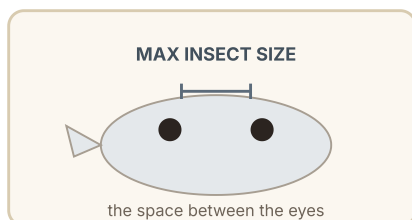
The signs are ordinary ones: diarrhea, fever, stomach cramps. That is why the link gets missed. Tell the doctor there is a reptile in the house, even if nobody remembers touching it, since enclosure surfaces are enough on their own. It matters most for a child under 5, an adult over 65, someone pregnant, or someone with a weakened immune system, in whom the infection can spread beyond the gut.

Diet & feeding by age

Leopard geckos are obligate insectivores: live insects and nothing else, no fruit or vegetable purees. Their nutrition is decided less by which insect you pick than by what that insect ate first (page 14) and what you dusted it with. Feed only when the warm side is at temperature: a gecko whose warm hide floor runs cool cannot digest properly, and a cool warm side is one of the most common reasons a gecko goes off food.

AGE	PREY SIZE	HOW OFTEN	HOW MUCH
Hatchlings and young geckos	About 1/4 in (6 mm)	Daily	A few insects at a time, or what it clears in about 15 minutes
Juveniles	Working up to slightly larger, never wider than the space between the eyes	Daily, or every 1 to 2 days	The same 15 minute rule
Adults, once the tail is as thick as the neck	No wider than the space between the eyes	Every 2 to 4 days	A few insects, adjusted to body condition (page 17)

The adult interval is not settled: anything from every other day to every 5 days once the tail has filled out is in use. Feed the gecko in front of you rather than the calendar.



Sizing and presentation

Prey too large to pass safely is one of the two usual causes of impaction, alongside loose substrate. Hatchlings take insects around 1/4 in (6 mm); after that, measure against the gecko rather than the calendar. Feeding tongs control the portion and keep insects from burrowing into the substrate.

When and how

- **Feed in the evening.** Leopard geckos hunt at dusk, so an evening feeding matches their natural activity better than a morning one.
- **Commercially raised insects only.** Never wild-caught, whatever the garden looks like.
- **Gut-load, then dust.** Every feeder spends 24 to 72 hours on a calcium-rich diet, then gets a light coat of powder right before it goes in (page 16).
- **Don't leave insects loose.** Feeder insects are not left unsupervised in the enclosure long-term; remove what isn't eaten.

FEED THE GECKO, NOT THE CALENDAR

The tail is the gecko's fat store and the best gauge of whether the schedule is right. A full tail and a steady weight mean it is working. A tail that thins over a couple of weeks means something is wrong, whatever the calendar says.

Feeder insects: staples, treats & never

Commercially raised only, gut-loaded first, and no wider than the space between the eyes.

INSECT	TIER	NOTES
Crickets	STAPLE	The usual staple. On their own they carry about 1 part calcium to 9 of phosphorus, so gut-loading and dusting are not optional.
Dubia roaches	STAPLE	A solid staple. A somewhat better calcium balance than crickets, but still well short of what a gecko needs without help.
Discoid roaches	STAPLE	Fill the same role as dubia roaches.
Black soldier fly larvae	STAPLE	The one common feeder that is not calcium-poor: calcium in the outer shell puts them close to or above 1 part calcium to 1 of phosphorus. Keep dusting anyway.
Mealworms	STAPLE	A common staple, but the worst calcium balance of the usual feeders, around 1 to 14. Gut-load and dust every time, and rotate with the roaches and crickets.
Hornworms (captive-bred only)	TREAT	Safe from a reptile supplier, where they are blue-green. An occasional treat.
Superworms	OCCASIONAL	Adults only. High in fat, and their size and hard outer shell are wrong for a juvenile at any size. Never a daily staple.
Waxworms	TREAT ONLY	Especially fatty. A couple of times a week at most.
Fireflies (lightning bugs)	NEVER	Genuinely toxic to leopard geckos.
Wild hornworms	NEVER	Bright green and toxic. Never off a garden plant; buy only from a reptile supplier.
Any wild-caught insect	NEVER	The pesticide and parasite risk isn't worth it, however clean the yard looks.

Why the calcium numbers matter

A gecko needs more calcium than phosphorus in its diet, roughly 2 parts to 1, or at the very least 1 to 1. Phosphorus competes with calcium for absorption in the gut, so a meal carrying plenty of calcium alongside far more phosphorus still leaves the gecko short. Analyses by Finke put plain crickets near 1 to 9 and mealworms near 1 to 14. A gecko fed nothing but plain feeder insects runs a calcium deficit at every meal, and sustained, that is the road to metabolic bone disease. Gut-loading and dusting close that gap between them.

Buying and keeping feeders

- Keep feeders in a ventilated container with room to move. Crowded crickets die fast and eat each other.
- Keep the gut-load diet and a calcium-fortified water gel in with them all the time, since insects drown in an open dish.
- Offer more than one staple. Some geckos lose interest in a repeated insect and pick back up on something different.
- Buying in bulk, or keeping a small dubia roach colony, brings the monthly feeder cost down significantly.

Gut-loading: what to feed the feeders

A gecko's nutrition is mostly decided a day or two before the insect is ever offered.

An insect is a container: whatever is in its gut when the gecko eats it gets eaten too. The effect is measured, not folklore: Klasing and colleagues (2000) found that crickets fed a diet of about 8 percent calcium for 48 hours had their whole-body calcium raised enough to bring the calcium to phosphorus balance into acceptable territory, and Allen and Oftedal (1989) showed that changing a cricket's diet changes what it delivers. The catch is that the diet has to be genuinely high in calcium, and that is where most gut-loading fails.

GUT-LOAD FOOD	TIER	NOTES
Dry commercial gut-load sold on its calcium content	STAPLE	Pick one sold on its calcium figure, not one that just says "gut-load" on the front. Labels vary a lot, one reason to dust as well.
Collard greens	STAPLE	Calcium-dense and low in oxalates, the combination that actually moves the balance.
Mustard greens, turnip greens	STAPLE	The same profile as collards.
Dandelion greens	STAPLE	Calcium-dense and low in oxalates.
Butternut squash	STAPLE	Calcium-dense, and insects take it readily.
Calcium-fortified water gel	STAPLE	Water and calcium in one. Insects drink even when they won't eat a dry diet. Never the only calcium source.
Carrot, potato, apple	OCCASIONAL	Keep insects alive and hydrated, which is worth doing, but as a calcium boost they do close to nothing. Not gut-loading on their own.
Spinach, beet greens, Swiss chard	NEVER	They contain calcium, but also oxalic acid, which binds that calcium into a form nothing can absorb. Self-defeating as a gut-load.

The timing window

24 to 72 hours on the diet before feeding, with 48 hours the most often cited. Too short and the insect has not eaten enough for its gut contents to matter. Longer stops helping, because gut-loading does not build up: an insect only carries what is in its gut right now. Insects start emptying their gut as soon as they come off the food, so feed them out within a few hours. With a standing colony, the simple approach is to leave the gut-load in with them all the time rather than treating it as a two-day event before each feeding.

Fresh food without the mold

Offer fresh produce in small amounts and remove it before it molds: mold in a warm feeder bin can kill a colony quickly. A gel cube diet handles food and moisture in one item without rotting, but treat it as what keeps the colony alive, not as the calcium source, and run a high-calcium dry diet alongside.

GUT-LOAD AND DUST, NOT ONE OR THE OTHER

A properly formulated high-calcium gut-load can, in principle, get an insect to an acceptable balance on its own. In practice, dust as well: you cannot see how much of the diet a given cricket actually ate, and powder on the outside does not empty out the way gut contents do.

Treats, refusals & the never-feed list

What to offer beyond the staples, and how to read a gecko that has stopped eating.

ITEM	VERDICT	NOTES
Waxworms, by tongs	TREAT ONLY	Especially fatty. A couple of times a week at most.
Superworms	OCCASIONAL	Adults only, and not a daily staple.
Hornworms, captive-bred	TREAT	Safe and blue-green from a reptile supplier.
Fruit, vegetables, purees	NEVER	An obligate insectivore. Its digestive system is not built for produce the way an omnivorous reptile's is.
Powdered fruit diets made for crested geckos	NEVER	A complete diet for a different species. Nothing substitutes for live insects here.

Why a leopard gecko stops eating

Most refusals are normal. Work through the list before assuming illness:

1. **A temperature too low.** One of the most common fixable causes. Read the warm hide floor first; it should be 88 to 92°F (31 to 33°C), with the cool side at 70 to 77°F (21 to 25°C).
2. **A shed.** Many geckos go off food for a day or two before, and sometimes after, a shed. Appetite returns once it is done.
3. **Brumation.** A natural winter slowdown cued by cooler temperatures and shorter days, with less activity and appetite and brief waking to drink (page 24).
4. **Stress.** A new enclosure, too much handling, or another gecko in the enclosure. It usually resolves once things settle.
5. **Breeding season.** Both sexes may eat less during courtship, and gravid females often eat less as they near laying.
6. **Prey boredom.** Some geckos lose interest in a repeated feeder and pick back up on something different.
7. **Illness or impaction.** Refusal alongside bloating, straining, loose stool, discoloration, poor shedding or sunken eyes points to a medical cause, not a cycle.

WHEN A REFUSAL BECOMES A VET VISIT

A healthy adult with a plump tail can typically go a couple of weeks without much concern, and the outer edge may be closer to a month. Rather than counting days, watch the tail: a noticeable thinning over a couple of weeks is the clearest warning sign, whatever day you're on. Go sooner for a refusal with lethargy beyond normal brumation, bloating, straining, discolored skin, poor or incomplete shedding, sunken eyes, or wrinkled, dry-looking skin.

THE NEVER-FEED LIST, IN ONE PLACE

Fireflies and lightning bugs · wild hornworms, the bright green ones · any wild-caught insect · fruit, vegetables and purees · powdered fruit diets made for crested geckos · anything wider than the space between the gecko's eyes · superworms for a juvenile · spinach, beet greens and Swiss chard as a gut-load.

Supplements: calcium, vitamin D3 & multivitamin

The dusting schedule and the heat are what a gecko's bones depend on, with UVB as the third leg.

SUPPLEMENT	JUVENILES & BREEDING FEMALES	ADULTS	NOTES
Plain calcium (no vitamin D3)	At most feedings	2 to 3× per week	The everyday dust, on the insects just before they go in.
Calcium with vitamin D3	A couple of times a week, alternated with the plain calcium, if the setup has no meaningful UVB		A gecko under a working UVB tube is making its own D3.
Multivitamin (reptile-specific)	About once a week	About once a week	A reptile multivitamin, not a general one.

The calcium dish

A shallow dish of plain calcium can stay in the enclosure at all times as an optional backup. It does not replace dusting: the schedule above is the plan, and the dish is a top-up the gecko can take on its own.

Dusting and gut-loading are two jobs

Gut-loading improves what is inside the insect; the powder adds calcium on the outside. Neither replaces the other. There is also a job gut-loading cannot do at all: vitamin D3. A gecko needs D3 to use calcium, from UVB on the skin or from its food, and gut-loading is a poor route for it. That is why, in a setup without UVB, the D3 dust is doing work nothing else covers.

- **Dust right before feeding.** Powder falls off, and an insect left dusted in a container for an hour has groomed much of it away.
- **Read the label.** The everyday dust is plain calcium with no vitamins added. Keep a D3 formula for its own slot.
- **Date the jars.** Replace opened powders by the date on the jar, and log the date on page 42.

IF THE SIGNS SAY CALCIUM, CHECK THE UVB AND HEAT

A gecko that is gut-loaded and dusted properly and still shows signs of calcium trouble, such as a soft jaw or tremors, usually has a light or heat problem rather than a powder problem. Leopard geckos appear to rely heavily on UVB and a correct warm side to process calcium, not on supplements alone.

FEMALES NEED CALCIUM WHETHER OR NOT A MALE IS AROUND

A female can make eggs with no male present, and making eggs draws on her calcium. A diet low in calcium is one of the causes of egg binding. Keep up the dusting for every adult female, with plain calcium at most feedings while she is breeding or laying.

Weight, the tail & body condition

The scale and the tail say whether the feeding schedule is working.

The numbers

Adult length	6 to 9 in (15 to 23 cm) nose to tail tip
Adult weight	Males 60 to 90 g; females lighter, some as light as 45 g
Adult tail	As thick as the neck, the point it moves onto the adult feeding schedule

Geckos vary, and a healthy one can sit at either end of those ranges. What matters is the trend: steady gain for a young gecko, a stable weight for an adult.

The tail is the fuel gauge

A leopard gecko stores its fat in its tail and lives off that reserve in lean times, through a brumation, or after a refused meal. A plump, thick tail means good condition. A thin one is the first sign of wasting, and a tail thinning toward a stick while the gecko still eats is the signature of cryptosporidiosis (page 27). Look at the tail every day; it shows trouble before almost anything else.

The weekly weigh-in

- Weigh once a week, at the same time of day and before a feeding, so a full stomach doesn't skew the reading.
- Use a kitchen scale that reads to single grams. In an animal this small, a slide of a few grams is invisible by eye and obvious on a scale. Write every reading on page 41.
- Watch the line, not the number. Through a suspected brumation, keep weighing at least every couple of weeks: a brumating gecko eats less but should not be losing significant body mass.

Body condition at a glance

UNDERWEIGHT

Hips, spine or ribs standing out, a sunken or narrow tail base, a body smaller than it should be for its age. Check the warm hide floor first, then see a vet rather than simply feeding more: parasites and cryptosporidiosis both look like this.

IDEAL

A rounded body with no projecting hips, spine or ribs, and a full tail as thick as the neck at its base. Weight stable week to week in an adult, climbing steadily in a young gecko.

WASTING

A tail thinning to a stick, weight falling, often with undigested food or foul-smelling stool, sometimes while the appetite stays normal. Keep it away from any other gecko and see a reptile vet promptly.

WHEN A WEIGHT CHANGE IS A VET VISIT

A tail that visibly thins over a couple of weeks. A thinning tail together with weight loss, at any time of year. Real weight loss during a suspected brumation. Any steady loss across the weekly log. Bring the log; a trend line tells a vet more than a single reading.

Handling & taming

Let it settle first

Wait one to two weeks after a new gecko comes home, and treat that as a benchmark rather than a countdown: don't start until it stops fleeing or hiding when you approach and is eating normally. A gecko under about 3 in (8 cm) long waits until it reaches roughly 6 in (15 cm) before regular handling, since hatchlings are fragile and easily injured. Settling is separate from quarantine, which runs for months if another reptile lives in the house (page 21).

Use the settling weeks to introduce your hand without touching: rest it flat in the enclosure for a few minutes each evening, when the gecko is up, and let it approach or tongue-flick your fingers on its own terms. A gecko that walks over to investigate a still hand is ready for the first pickup.

How to pick one up

Use a two-handed scoop: one hand goes under the body to support the hips and the base of the tail, and the gecko walks onto your hand rather than being grabbed from above. Support the feet, body and tail together in an open palm, and let it walk hand to hand rather than holding it still. Stay low to the ground: leopard geckos jump and dart without warning, and a fall from height is a real injury for an animal this small.

NEVER BY THE TAIL

A leopard gecko can drop its tail, called caudal autotomy, when it is grabbed by the tail, badly frightened, handled roughly or attacked by another gecko, and it can come off with very little pressure. The tail is its fat store, so losing it has a real cost. It regrows in about 30 days, but the new one is built on cartilage instead of bone and comes back shorter, thicker and different-looking (page 31).

How often and how long

Once settled, start with 5-minute sessions every other day and build gradually toward 10 to 20 minutes. Cap a young gecko at a few short sessions a week. Evening works best, close to its natural active time, but not right after it wakes. It doesn't need daily handling. Never handle during a shed, and skip handling for about 48 hours after a large meal: jostling a full gecko can make it bring the meal back up.

PUT IT BACK WHEN YOU SEE

Chirping or squeaking · screaming, more common in juveniles · a slow side-to-side tail swish, usually with an arched back and the gecko up on its toes · fast retreats or attempts to hide · refusing food afterward. A bark alone isn't necessarily distress and can just mean hunger; the others end the session.

Who should hold it

Children under 5 should not handle the gecko or touch its enclosure, because a healthy-looking gecko can still carry *Salmonella* (page 11). Older children handle it with an adult supervising. Wash hands with soap before and after every session.

Body language: tail, sounds & posture

The tail carries most of what a leopard gecko is saying. Read it before you reach in.

YOU SEE	IT USUALLY MEANS	DO
A slow, deliberate tail swish, back arched, legs straightened, up on its toes	A warning: leave it alone	Close the enclosure and try again once the posture relaxes
A rapid tail flick or rattle	Excitement, usually at prey, or in an adult male at a potential mate	Let it finish hunting; handle it later
Chirping or squeaking, while held or when startled	It has had enough, or it was startled	Put it back
Screaming, most often a juvenile	Stress	Put it back and give it time
Barking	Not necessarily distress; it can just mean hungry	Check when it last ate
Walking over to sniff or tongue-flick a still hand	Curiosity, and readiness for handling	Try the first scoop (page 18)
Hidden most of the day, out at dusk and dawn	Normal: it is crepuscular	Feed and handle in the evening
Pacing or glass surfing away from the warm side, gaping, refusing the warm hide, flattening into the coolest spot	Too hot, with heat it cannot escape	Read the warm hide floor and fix the heat
Darkened skin, wobbly, limp	Overheating	An emergency: somewhere cooler, heater off, call a vet
Sitting in the water dish for unusually long spells	Possibly mites, which a reptile may try to drown	Look closely around the eyes and under the chin
Eating less, hiding more, quieter as the days shorten	Possible brumation, the winter slowdown	Weigh it and watch the tail (page 24)
An adult female restless, digging, belly bulging, eating less	Gravid: carrying eggs, with or without a male	Lay box in now (page 22)

Too cold looks quiet

Too hot is loud. Too cold is quiet: it shows slowly, as a gecko that eats less and moves less, long before anything shows on the animal. That is why a refused meal sends you to the infrared thermometer before anything else. Open-mouth breathing, discharge from the nose or labored breathing in a cool enclosure is a respiratory problem, not a mood (page 30).

QUICK DIFFERENTIATOR

Brumation looks like a gecko that is quiet, hiding and off food but holding its weight and its tail. Illness looks like the same behavior with a thinning tail, weight loss, sunken eyes or real lethargy. The scale is what separates them.

Enrichment & common mistakes

Mistakes that cause the most harm

- One hide instead of three. It is not enrichment with a gap; it is a missing foundation.
- Skipping the humid hide, or raising the humidity of the whole enclosure instead of using one.
- A heat mat with no thermostat, or a probe anywhere but the warm hide floor.
- The old 20 gallon long tank in place of the 36×18×18 in (91×46×46 cm) minimum.
- Calcium sand, including sand sold as an enrichment substrate.
- Lifting a gecko by the tail during "play," or treating a shed as handling time.
- Wild-caught insects or fireflies as a "natural" treat.
- Assuming a wheel or a toy substitutes for hides and hunting.

What the research says

Leopard geckos are unusual in having real studies behind their enrichment. Bashaw and colleagues (2016) tested five types of enrichment: thermal and feeding enrichment produced the strongest responses, and mirror-based visual enrichment the weakest. Rickman and colleagues (2025) found better welfare in enriched housing than in sparse setups, and geckos given the choice consistently picked the enriched side. Enrichment comes after correct belly heat and three hides, never instead of them. In this order:

1. Belly heat and the three hides

The foundation. A warm dry hide, a cool dry hide and a maintained humid hide let a gecko choose its microclimate while staying hidden. That choice only exists if the gradient under the hides is right.

2. Feeding and hunting

The highest-value enrichment after heat. Tong-feeding, short supervised insect hunts inside the enclosure, a rolling dispenser that drops insects as it is nudged, and simple timed-release prey containers all beat a static bowl. Offering small insects in a more natural way has been linked to lasting increases in activity. Feed in the evening, when it hunts.

3. Low climbing and cover

Low sturdy branches, cork flats or tubes and stacked slate give somewhere to climb and rest off the ground, plus cover that still lets it move through the space. Height matters less than floor space.

4. A dig box

A single box to dig in satisfies the digging some geckos do, especially some females, without turning the whole floor into loose substrate. Paper towel or tile stays the main floor. Never fill it with calcium sand.

5. Light novelty

A new piece of cork, a mild rearrangement. Some value, genuinely last on the list, and no substitute for anything above it.

What barely registers

Mirrors and other visual novelty were the weakest result in the research, and a wheel or a toy does not stand in for hides and hunting. The answer is more of items 1 to 4, not novelty for its own sake.

Choosing a gecko, quarantine & the first vet visit

Captive-bred, checked over in good light, and in front of a vet within 72 hours.

A breeder who hatched the gecko can tell you its hatch date, what it eats, and when it last ate and shed. A pet store or breeder is worth buying from only if its enclosures are clean and large enough. A rescue or an expo animal often comes with no reliable history: not a reason to refuse it, but a reason to run the full quarantine below. A seller who cannot say whether an animal was bred or caught has answered the question.

Look before you pay

CHECK	HEALTHY	WALK AWAY OR ASK MORE
Tail	Thick and plump: the fat store	A thin tail, the first sign of wasting disease
Body	Rounded, no projecting hips, spine or ribs	Visible bones, a sunken tail base, small for its age
Eyes	Clear and clean	Swollen, sunken, cloudy outside a shed, or crusted
Nose, breathing	Dry nostrils, quiet breathing	Bubbles, mucus, wheezing, open-mouth breathing
Vent (the opening under the tail)	Clean	Stained, pasted with droppings, or swollen
Skin	Intact, no old shed left on	Wounds, or rings of old shed on toes and tail tip
Behavior	Bright, alert, tongue-flicking, a good appetite	Limp, unresponsive, or unable to hold on

Ask the seller: the hatch date; whether it was captive-bred; what it eats and when it last ate; whether it has been treated for anything; and whether it has had a fecal test (a stool check for parasites). A good breeder answers without looking anything up.

Quarantine: 3 to 6 months

If another reptile already lives in the house, keep the new gecko completely apart for 3 to 6 months. 60 to 90 days is a floor, not a target: apparently healthy geckos can carry cryptosporidiosis, which has no cure, without showing a sign. With no other reptile, skip the separation but not the vet visit.

- **A separate room.** Two enclosures on one shelf share the same air.
- **Paper towel, a hide and a water dish.** Paper shows stool, mites and a regurgitated meal at a glance.
- **Separate tools, and the new gecko last.** Its own tongs, dishes and feeder tub, cared for after every other animal, with hands washed in between.
- **Restart the clock** if anything flags: a positive fecal test, a regurgitated meal, unexplained weight loss. At the end, disinfect everything that touched it.

THE FIRST VET VISIT

Book a reptile vet before the gecko comes home (finding one is on page 26), for within 72 hours of arrival and no later than two weeks. The first exam records a weight and a look over the whole animal, and checks a fresh stool sample for parasites, so bring one. Ask for the cryptosporidiosis PCR (polymerase chain reaction) test by name: a lab test that finds a germ's genetic material in the sample. That visit is the baseline every later one is measured against.

Sexing & females that lay

Every female owner needs this page, whether or not a male has ever been in the house.

Telling male from female

Look at the underside, near the vent. It only works on a maturing gecko, so treat a hatchling sold as male or female as a guess and check again as it grows. **A male** shows a V-shaped row of enlarged preanal pores (waxy pores just in front of the vent) and a more bulbous tail base where the hemipenes, the male's paired reproductive organs, sit as two hemipenial bulges. **A female** lacks the enlarged pore row and has a flatter tail base. Never hold the gecko by the tail to look.

Sex is set in the egg. Rhen and colleagues (2011) bred leopard geckos over 13 generations: eggs incubated at 86°F (30°C) hatched mostly female and eggs at 90.5°F (32.5°C) mostly male, so a breeder who records incubation temperatures can say which way a clutch was likely to go. A temperature is a probability, not a sexing.

Females lay without a male

A female leopard gecko can produce eggs with no male in the enclosure. Those eggs are infertile, but making them is the same physical work, and most reptiles brought to a vet for egg binding are single females that have never met a male. So every adult female needs a place to lay, and calcium in her diet whether or not she is kept with a male (page 16).

Signs she is gravid

- A bulging belly and weight gain.
- Restlessness and more digging.
- A smaller appetite.

The lay box

A covered container large enough for her to turn around in but small enough to feel enclosed, filled with damp sphagnum moss she can dig into. Damp but not waterlogged, holding about 70 to 80% humidity inside the box. Place it in a quiet corner where she feels secure. A female with nowhere suitable to lay may hold her eggs, which is how a normal clutch turns into egg binding.

THE CHEAPEST PREVENTION THERE IS

A lay box ready for any adult female, correct temperatures, humidity and UVB, and a calcium-rich diet cover most of what causes egg binding. The page after this one is the one to know before you ever need it.

Egg binding

Dystocia: a female who cannot pass the eggs she is carrying. A same-day emergency.

Egg binding can be fatal without intervention. A gravid female and one in trouble share some signs at first: both are restless and both eat less. The difference is what comes next. A female in trouble strains with nothing produced, turns weak and depressed, and stops eating altogether with a distended belly and no eggs laid.

SIGN	NORMAL GRAVID FEMALE	EGG BINDING
Appetite	Smaller	Lost, in a female that has not laid
Activity	Restless, digging	Restless, then weak and depressed
Belly	Bulging, with weight gain	Distended
Effort	Digs, then lays	Straining with nothing produced

CALL A REPTILE VET THE SAME DAY IF

She is straining and producing nothing · she is weak, depressed or restless without laying · her belly is distended and she has stopped eating without having laid · any tissue is protruding from her vent, which is an emergency on its own.

What causes it

- **Nowhere to lay:** no lay box, or one she doesn't feel secure in.
- **Husbandry:** temperatures too hot or too cold, the wrong humidity, missing UVB.
- **Diet:** too little calcium or protein, and dehydration.

What the vet does

A vet may start with fluids, calcium and hormone injections to help her lay, and surgery follows if those fail. Because it can be fatal, it is a same-day visit, not a wait-and-see.

PREVENTION, IN ONE LINE

Every adult female gets a lay box, correct heat and humidity, and calcium dusting, whether or not a male has ever been near her. Most egg binding starts with one of those missing.

The shed cycle & brumation

Two normal cycles that look like illness to a new owner, and how to tell them apart from the real thing.

The shed cycle

Reptiles shed their skin, a process called ecdysis, to grow and to clear parasites and old, damaged skin. Before a shed, the gecko's colors go dull and its skin looks dry, and it often hides more and refuses food for a day or two. Then it rubs against rough surfaces to start the peel and works its way out of the old skin.

- **Humidity, in the right place.** The humid hide is what lets old skin come away in one piece. Keep it damp, especially when a shed is due, rather than making the whole enclosure damper.
- **Something rough.** Cork bark, slate and branches give it something to rub against to start the peel.
- **Hands off.** No handling during a shed. Expect a refused meal or two; appetite returns once the shed is done.
- **Check afterward.** The toes, the tail tip and around the eyes, in good light, after every shed. Retained skin collects in those three places, and a ring of it can cut off the blood supply (page 31).

SHED WEEK ROUTINE

Top up the humid hide. Leave the gecko alone. Don't worry about a refused feeding. The day after, check toes, tail tip and eyes. Anything still attached gets a soak, never a fingernail.

Brumation

Brumation is a reptile's natural winter slowdown, cued by cooler temperatures and shorter days. You don't have to induce it. A brumating gecko is less active, eats less or stops eating, and wakes briefly to drink. It lives off the fat in its tail meanwhile, which is exactly why the tail and the scale tell you whether it is going well.

Weigh it	At least every couple of weeks, in grams. A brumating gecko eats less but should not be losing significant body mass.
Watch the tail	It should stay full. A tail that visibly thins is not brumation.
Keep water out	Fresh water in the dish every day: it wakes briefly to drink.

NOT BRUMATION: CALL THE VET IF

The tail is visibly thinning · weight is falling · it is lethargic beyond normal brumation behavior · there is bloating, straining, discolored skin, sunken eyes or wrinkled, dry-looking skin. A slowdown with any of these is illness until a vet says otherwise.

Is a leopard gecko legal where you live?

Legal without a permit almost everywhere in the United States, with a short list of exceptions.

This page covers all 50 states, the District of Columbia and New York City, which has its own rules: 52 jurisdictions. A pet leopard gecko is **legal with no permit in 46, barred in 2, permit-only in 2, and legal on a condition in 2.**

WHERE	STATUS	THE RULE
Hawaii	Barred	Named by species on the Restricted Animal List, Part A (Hawaii Administrative Rules § 4-71-6.5), which is for research and exhibition only. No route for a household.
District of Columbia	Barred	District of Columbia Code § 8-1808(j)(1) bars keeping wildlife except seven listed groups, and no lizard fits any of them. No permit sits behind the bar.
New Jersey	Permit	Named on the permit list as "Geckos - Family Gekkonidae other than Tokay Gecko" (New Jersey Administrative Code 7:25-4.3(a)3.xiv). The permit is the individual hobby permit (7:25-4.6(a)1).
West Virginia	Permit	West Virginia Code § 20-2-13 bars bringing in or keeping live wildlife from out of state without an importation permit from the director, captive-bred or not. No application is published: ask the state in writing before the gecko crosses the line, and keep the reply with its purchase records.
Maine	Conditional	Cleared on the Unrestricted Species List as "all species in infraorder Gekkota," under the heading "Captive Bred Origin Only." Close to a formality, since captive-bred is what the trade sells.
Minnesota	Conditional	Minnesota Statutes § 97A.401, subdivision 8: a lizard from a permitted breeder, or held before 1 August 2021, may be kept as a pet. The breeder permit is not visibly in place yet, so ask the state wildlife agency how to meet the condition.

Legal with no permit: 46 jurisdictions

Alabama, Alaska, Arizona, Arkansas, California, Colorado, Connecticut, Delaware, Florida, Georgia, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maryland, Massachusetts, Michigan, Mississippi, Missouri, Montana, Nebraska, Nevada, New Hampshire, New Mexico, New York State, New York City, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, South Dakota, Tennessee, Texas, Utah, Vermont, Virginia, Washington, Wisconsin and Wyoming.

In eight of them, Colorado, Illinois, Maryland, Michigan, Montana, Nevada, Oregon and Washington, the state lets cities or counties be stricter, so check your local ordinance (a city or county law) too. Bringing a gecko into Michigan calls for a certificate of veterinary inspection.

NO FEDERAL OR TREATY RULE REACHES IT

The leopard gecko is rated Least Concern, is not on Appendix I of CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora, the treaty tier for the most endangered species), and appears on neither federal wildlife list. Leopard geckos were being bred in the United States by the early 1970s, and the trade is essentially all captive-bred.

NOT LEGAL ADVICE

State legal status as of October 2026. This is a summary of state rules, read on 1 October 2026 (Maine on 5 September 2026), and not legal advice. Rules change. Before you buy or move, check with your state wildlife agency and with your city or county.

Red flags & finding a vet

What to watch for, what to tell the vet, and how to find one before you need one.

CONTACT A REPTILE VET PROMPTLY IF YOU SEE

A tail thinning to a "stick," especially with weight loss · a soft or deformed jaw · tremors · any breathing difficulty, or open-mouth breathing · drooling or pus around the mouth · appetite loss that persists more than a few days alongside any other sign here, or a tail that thins while off food · a bloated belly with no normal bowel movements · straining with nothing produced · blood in the stool · any tissue protruding from the vent · retained shed that is swollen, darkened, or won't clear from the eyes or toes · darkened skin, wobbliness or limpness in the heat · a female straining or weak without laying.

What to tell the vet

- ☐ The warm hide floor reading from an infrared thermometer, not the thermostat's setting, and the cool side
- ☐ Substrate, and whether loose substrate has ever been used
- ☐ Which powders, how often, and whether UVB is running
- ☐ When symptoms started and whether they're getting worse
- ☐ Last normal bowel movement, and what it looked like
- ☐ Weight trend from the Owner Log, and a fresh stool sample in a sealed bag if you can

Why the heat comes up first

A leopard gecko cannot make its own body heat, so the warm hide floor is not a comfort setting: it powers digestion and the immune response. A floor a few degrees low brings refused meals, slow stools, a gecko that hides all day, and in time a real infection, each looking like a different problem. That is why a vet asks for the number before anything else.

Finding a reptile vet

Many dog and cat practices do not see reptiles, so find one early. The Association of Reptile and Amphibian Veterinarians keeps a find-a-vet directory of its members, and a member is the place to start. Ask a clinic how many reptiles it sees, whether it treats leopard geckos routinely, whether it runs fecal tests in house, and who covers emergencies after hours, then write the number on page 34. Book a wellness exam once a year even when the gecko seems perfectly healthy: several of the conditions in this section, cryptosporidiosis above all, can be present without obvious signs until they are advanced.

Normal or concerning

NORMAL	CONCERNING
Dull, dry-looking skin, more hiding and a skipped meal or two before a shed	Shed left stuck on the toes, tail tip or around the eyes after the shed
Less appetite and activity in cooler months, with the tail full and the weight steady	Less appetite with a tail that is visibly thinning, at any time of year
Hidden most of the day, out and about at dusk and dawn	Sluggish and hard to rouse in its evening active time
A bark or chirp when it wants food or space	Clicking, wheezing or open-mouth breathing at rest

Cryptosporidiosis: stick tail disease

The one condition every leopard gecko owner should be able to recognize. It is severe, highly contagious, and has no cure.

What it is: an infection with the parasite *Cryptosporidium*. In leopard geckos it commonly causes clinical disease and wasting, though not every carrier shows signs right away, and apparently healthy carriers spread it. One study gives a sense of scale: Nipithakul and colleagues (2025) screened the geckos at one Thai breeding facility with a PCR test and found *Cryptosporidium varanii* in 51% of them. It earned its common name from the sign that gives it away: a healthy tail is thick with stored fat, and this disease melts it down to a stick.

Signs

- A tail that thins to a "stick," with no fat reserve left.
- Weight loss and an emaciated look overall.
- Loss of appetite.
- Undigested food in the droppings, and foul-smelling stool.
- A bloated or distended belly.

Diagnosis and treatment

A reptile vet diagnoses it with a fecal PCR test. False negatives happen, so one clear test does not fully rule it out. The main treatment drug, paromomycin, manages the condition rather than curing it. Mortality for geckos with symptoms runs around 50%, and a gecko that hasn't responded well to treatment within a month generally has a poor outlook.

A THINNING TAIL IS CRYPTOSPORIDIOSIS UNTIL A TEST SAYS OTHERWISE

See a reptile vet promptly for any tail thinning toward a stick, especially with weight loss. Keep that gecko away from every other reptile, with its own tools, cared for last, with hands washed in between, until the vet has answered the question.

Prevention is the only real control

With no cure and carriers that look healthy, prevention is about keeping it out. Quarantine any new gecko for 3 to 6 months in a separate room before it shares a room, a tool or a hand with an existing animal (page 21). Ask a breeder whether the gecko has had a fecal test before you buy. Bring a fresh stool sample to the first vet visit and ask for the cryptosporidiosis PCR test by name, then keep up the yearly wellness exam.

WHY THE SCALE MATTERS MOST HERE

A gecko losing weight while it still eats is the pattern to catch. The weekly weigh-in shows it weeks before the tail looks obviously thin, and it is the difference between this disease and a normal winter slowdown.

Metabolic bone disease

Common in leopard geckos, and largely preventable once heat and calcium are handled correctly.

What it is: metabolic bone disease (MBD) is a weakening of the skeleton from too little usable calcium. **Why it is common in this species:** leopard geckos are so often kept without UVB, without proper calcium supplementation, or without enough heat, and all three contribute. This species appears to rely heavily on UVB and a correct warm side to process calcium, not on supplements alone, which is part of why low-level UVB is worth providing even though it is technically optional.

Signs

WHERE	SIGNS
Jaw	A soft or deformed jaw, sometimes called "rubber jaw"; difficulty eating
Limbs and bones	Twisted or bent limbs; fractures from minor knocks or falls
Nerves and muscles	Tremors
Whole body	Lethargy and weight loss

Response: see a reptile vet promptly. Any softness in the jaw, any tremor, or any new difficulty getting around is an appointment, not a wait-and-see.

WHILE YOU WAIT FOR THE APPOINTMENT

Handle the gecko extremely gently, and as little as possible. Take out any climbing decor: weakened bones can fracture in a fall that would not normally matter.

PREVENTION BEATS TREATMENT

A warm hide floor at 88 to 92°F (31 to 33°C) on a thermostat, gut-loaded and dusted insects on the schedule (page 16), and low-output UVB replaced on time (page 9) prevent nearly all of it. A cool warm side is one of the usual ways it starts, and the easiest one to miss.

Why the three legs

Calcium from the food is only useful if the body can absorb it, and that takes vitamin D3, which a leopard gecko makes in its skin under UVB or takes in from a D3 dust. Digesting the meal in the first place takes heat from the floor. Take away any one of the three, calcium, D3 or heat, and the bones pay for it.

Impaction, dehydration & prolapse

Three problems that often travel together: a dry, cold or blocked gut, and the straining that follows.

Impaction

What it is: a blockage in the digestive tract. **Cause:** usually swallowed loose substrate, most often sand, or prey too large to pass safely. A warm side too cool to digest, dehydration and parasites all make it likelier. **Signs:** a bloated belly, loss of appetite, weight loss, and no normal bowel movements. **Response:** warm baths and gentle belly massage help mild cases; a true blockage may need surgery. That is the reason paper towel and tile are the safer floor, and the reason prey is measured against the space between the eyes rather than guessed.

WHEN IT IS A VET VISIT

A bloated belly with no normal droppings, straining with nothing produced, or a gap in droppings well beyond your gecko's own normal rhythm. Know that rhythm from the log, so a gap means something.

Dehydration

A desert species still drinks, and dehydration builds quietly. The urates usually show it first.

SIGNS OF DEHYDRATION

Wrinkled, dry-looking skin outside a shed, sunken eyes, and urates that are yellow, orange or gritty instead of smooth and white. Give it a bigger, easier-to-reach water dish and a properly damp humid hide. A gecko that stays sunken and wrinkled, or that is also off food and losing weight, is a vet visit.

Prolapse

Pink or red tissue protruding from the vent. It is always an emergency: get to a vet immediately. On the way, keep the tissue moist with clean water and never try to push it back in. It is a sign of an underlying problem rather than a condition on its own, and straining, from a blockage or from a female trying to lay, is one of the usual routes to it.

Respiratory infection, parasites & mites

One that comes from the enclosure, and two that usually come in with a new animal.

Respiratory infection

What it is: a bacterial infection of the airways. **Cause:** in a leopard gecko it almost always starts with an enclosure that is too cool, too damp, or both, and more than a day or two well below the gecko's comfortable range is the textbook setup for it. **Signs:** open-mouth breathing, bubbly saliva, clogged nostrils, lethargy, loss of appetite and weight loss. **Response:** it does not clear on its own. A reptile vet prescribes antibiotics, usually with a probiotic, and a sick gecko may need extra fluids and help with feeding while it recovers. At home, fix what caused it: bring the warm side back up to temperature, and find the source of excess moisture, such as a humid hide that has soaked the substrate around it. That is also why raising the humidity of the whole enclosure to fix a shedding problem is the wrong move.

Internal parasites

Gut parasites, coccidia (single-celled gut parasites), other protozoa and worms, show no outward sign in many geckos, which is why a microscope check of a fresh stool sample is part of every first visit and every yearly one. When they do show, repeated watery stool is the classic sign, often with weight loss and poor condition. Bring a fresh sample in a sealed bag. Treatment depends on which parasite the fecal test finds, which is why a test comes before any medicine.

Mites

Tiny black, brown, reddish or orange flecks on the head, neck or belly, especially around the eyes. A reptile with mites often sits in its water dish for unusually long spells, as if trying to drown them. To check, gently rub the gecko while it stands over a sheet of white paper and look for anything that falls off. Mites arrive with a new animal or on shared equipment, spread fast, and can carry other infections, so a confirmed case is a vet visit and a reason to check every reptile in the house.

THE COMMON THREAD

A correctly heated enclosure that is dry everywhere except inside the humid hide prevents most respiratory infections. A full quarantine on paper towel, separate tools and a fecal test catch most parasites and mites before they reach anything else you keep (page 21).

Stuck shed, tail loss & burns

Skin and tail problems a look after every shed catches early.

Stuck shed (dysecdysis)

Old skin that stays on after a shed collects in the same three places: around the eyes, on the toes, and at the tail tip. A ring of it can tighten around a toe or tail tip and cut off its blood supply badly enough to cause tissue death and lost toes, and the damage builds over several sheds. In leopard geckos the usual causes are low humidity, which a humid hide fixes, or too little vitamin A. Six things in all can cause a bad shed: low humidity, skin parasites, poor nutrition, infection, nothing rough to rub against, and low thyroid function. Repeated bad sheds in a correct setup are a vet conversation, not a humidity problem.

- **Never peel dry, stuck shed.** Soak first.
- **The soak:** shallow water at enclosure temperature, chin-deep at most, for 30 minutes, supervised throughout. Then ease the loosened skin off a toe or tail tip with a damp cotton swab.
- **Eyes are a vet job.** Retained shed around the eyes that won't clear, or any shed that is swollen or darkened, goes to a reptile vet.

Tail loss

A dropped tail is a stress response: a gecko grabbed by the tail, badly frightened, handled roughly or attacked by another gecko. If it happens:

- Move the gecko onto clean paper towel straight away, and keep the enclosure especially clean while the stump heals.
- No handling for 1 to 2 weeks.
- Watch the stump for swelling, discharge, discoloration or a foul smell. Any of those means a vet.

The tail regrows in about 30 days, built on cartilage instead of bone, shorter, thicker and different-looking. The real cost is the fat it carried, so keep a closer eye on the weight log for the next few months.

Burns

Unregulated heat is one of the most common causes of thermal burns in captive leopard geckos: a heat mat with no thermostat, a thermostat whose probe has moved, a hot rock, or an overhead lamp it can reach. Burns show as blistered, blackened or peeling skin. Turn the heat source off, fix the regulation, and get anything past a small red patch seen by a vet, because reptile burns often look worse over the following days rather than better.

Reading poop, urates & hydration

Stool is the earliest free health check you get. Look at it before you clean it up.

Urates are the white part of a reptile's droppings. Instead of liquid urine, the body turns nitrogen waste into uric acid, made in the liver from protein digestion, and passes it as a white to cream, chalky or paste-like portion alongside the stool. Its color and texture are the fastest daily read on hydration you have.

IT LOOKS LIKE	USUALLY MEANS	DO
A firm, well-defined dropping, often with visible bits of insect shell, and a smooth white to cream urate	Normal for an insect eater. Bits of hard insect shell are expected	Nothing. Note it in the log.
Urate yellow, orange, or gritty and crystal-like	Dehydration at least, and left uncorrected a urate or kidney problem	Bigger, easier water access and a damp humid hide. Vet if it persists: the same uric acid chemistry causes gout.
Watery or unformed for more than one dropping in a row	Parasites, especially single-celled ones	Vet with a fresh sample in a sealed bag.
Undigested food, foul-smelling stool	Possible cryptosporidiosis, or a warm side too cool to digest	Read the warm hide floor today; vet, especially if the tail is thinning (page 27).
Bright red blood, or dark and tarry	Fresh bleeding near the vent, or digested blood from higher up	Vet the same day.
An unusually strong smell for your gecko	A red flag, especially with other changes	Vet, sooner if it comes with lethargy or a skipped meal.
Straining, with nothing passed	A blockage, a problem at the vent, or in a female, eggs	Vet promptly; straining can end in a prolapse.
Nothing for well beyond your gecko's normal gap, with a bloated belly	Possible impaction	Vet promptly (page 29).

How often is normal

Most adult geckos go every few days to about once a week, and young ones more often, since they eat more often. The number that matters is your own gecko's baseline. A real change from it, in how often, how it looks, or how it smells, is the signal, not any fixed count.

Hydration

- A shallow dish of fresh dechlorinated water, changed daily, that the gecko can climb in and out of easily.
- The humid hide is hydration equipment as well as shedding equipment. Keep it damp all year.
- Dry, hard or gritty urate is the signal to act: more water access for a ground-dwelling species like this one, not more misting of the enclosure.

Setup checklist & targets

Essential setup checklist

- ☐ 36×18×18 in (91×46×46 cm) enclosure, front-opening, escape-proof screen top (page 6)
- ☐ Under-tank heat mat under the warm hide
- ☐ On/off thermostat, probe on the warm hide floor on a clip or suction cup, never tape (page 8)
- ☐ Infrared thermometer for the warm hide floor
- ☐ Digital probe thermometers and a hygrometer
- ☐ Plug-in timer for the lights: 12 hours on, 12 off
- ☐ Optional low-output T5 HO UVB tube, set for a UVI of 0.6 to 1.4 (page 9)
- ☐ Three snug hides: warm dry, cool dry, and a humid hide with a damp lining (page 10)
- ☐ Paper towel or tile for the floor
- ☐ Shallow water dish and a small calcium dish
- ☐ Plain calcium, calcium with vitamin D3, and a reptile multivitamin (page 16)
- ☐ Feeding tongs and a kitchen scale that reads in grams
- ☐ A feeder insect supplier and a high-calcium gut-load diet (page 14)
- ☐ A plastic tub for cleaning gear, never the kitchen sink (page 11)

Temperature & climate targets

TARGET	°F	°C
Warm hide floor (belly heat)	88 to 92	31 to 33
Warm side, ambient air	80 to 84	27 to 29
Cool side	70 to 77	21 to 25
Night, whole enclosure	the 70s; never below 65	21 to 26; never below 18
Humidity	30 to 40% in the enclosure, 70 to 80% inside the humid hide	
UVI at the basking spot	0.6 to 1.4 (0.5 to 0.7 or lower for albino and pale morphs)	
Light cycle	12 hours on, 12 hours off, on a timer. No colored night bulbs	

Before your gecko comes home

- ☐ Enclosure set up and running, warm hide floor read with the infrared thermometer, not by feel
- ☐ Thermostat watched cutting the mat as the probe warms
- ☐ Humid hide lined and damp, all three hides in place
- ☐ UVB tube mounted at the right height, its date logged (page 42)
- ☐ Feeder insects and gut-load on hand before day one
- ☐ Reptile vet found and the first exam booked for within 72 hours (page 21)

Emergency & quick targets

CALL A REPTILE VET NOW IF

The tail is thinning to a stick, especially with weight loss · the jaw is soft or crooked, or there are tremors · breathing is open-mouthed, bubbly or labored · tissue is protruding from the vent · a female is straining or weak without laying · a bloated belly with no droppings, or straining with nothing produced · blood in the stool · blistered or blackened skin · darkened skin, wobbliness or limpness in the heat · the gecko is limp or cannot be roused

READING	°F	°C
Warm hide floor (belly heat)	88 to 92	31 to 33
Warm side, ambient air	80 to 84	27 to 29
Cool side	70 to 77	21 to 25
Night: normal in the 70s; act below (outage floor)	65	18
Humidity	30 to 40% in the enclosure, 70 to 80% inside the humid hide	
UVI at the basking spot	0.6 to 1.4 (0.5 to 0.7 or lower for pale morphs)	
UVB tube replacement	Every 6 to 12 months, by the model's rated life	
Calcium	Plain at most feedings for juveniles and breeding females, 2 to 3× a week for adults · D3 version a couple of times a week with no UVB · multivitamin about once a week	

Your numbers

Reptile vet _____

UVB tube installed _____

24-hour exotic emergency clinic _____

Thermostat last checked _____

Gecko's name and hatch date _____

Last vet visit _____

HAVE THIS READY WHEN YOU CALL

The warm hide floor reading from the infrared thermometer, and the cool side. Current weight in grams and the trend from the log. Substrate. Which powders, how often, and whether UVB is running. When it last ate and last passed a dropping, and what it looked like. Transport in a ventilated tub with a towel, kept warm (page 39).

Budget & shopping list

United States retail ranges, last checked September 2026. Prices move; treat this as a snapshot, not a quote.

INITIAL SETUP	TYPICAL COST
Enclosure, 36×18×18 in (91×46×46 cm) minimum	\$50 to \$250
Under-tank heat mat and thermostat	\$50 to \$70
Low-output UVB fixture and tube (optional, recommended)	\$20 to \$50
Digital thermometer and hygrometer	\$15 to \$40
Substrate: paper towel or tile	\$10 to \$25
Three hides and a water dish	\$30 to \$60
Equipment, the low to the high end of every row	\$175 to \$495
Equipment for most first-time setups	\$250 to \$400
First vet exam and fecal test	\$150 to \$215
All-in for most setups, before the gecko arrives	\$400 to \$615

The gecko itself is extra: \$20 to \$45 for a wild type from a pet store, \$50 to \$150 from a breeder even for a basic morph, \$75 to \$300 for popular morphs, \$500 to \$3,500 or more for rare ones, plus \$35 to \$100 for live shipping if you buy online. Not priced above, but on the list: an infrared thermometer, a plug-in timer, feeding tongs and a kitchen scale.

MONTHLY ONGOING	TYPICAL COST
Feeder insects	\$10 to \$30
Calcium and vitamin supplements	\$2 to \$10
Electricity	under \$15
Substrate and miscellaneous	\$4 to \$20
UVB tube replacement, if used, averaged	\$4 to \$7
Every row at its range, electricity counted from \$0	\$20 to \$82
Monthly ongoing for most owners	\$20 to \$50

Add the yearly wellness exam, \$85 to \$105. Most owners spend \$250 to \$700 a year in all. Buying feeders in bulk, or keeping a small dubia roach colony, is the biggest lever on that number.

PLAN FOR, DON'T COUNT ON AVOIDING

A sick or emergency visit runs \$100 to \$800 or more, and diagnostics or surgery climb from there. A small emergency fund set aside matters as much as any item on this list.

WHERE NOT TO CUT CORNERS

The thermostat and the enclosure size. The thermostat is what stops the mat from overheating and burning the gecko; save on decor before you save there.

First 30 days

Day 1

- ☐ Weigh the gecko in grams and photograph it and the setup for a baseline
- ☐ Read the warm hide floor with the infrared thermometer and the cool side with a probe, not "feels warm"
- ☐ First exam with a reptile vet within 72 hours, with a fresh stool sample and the cryptosporidiosis PCR test asked for by name (page 21)
- ☐ If another reptile lives here, set up the separate quarantine room and its own tools
- ☐ Fill in the emergency card (page 34)
- ☐ Start the equipment dates (page 42)

Week 1

- ☐ No handling. Fresh water daily, food offered in the evening, uneaten insects removed
- ☐ A hand resting flat in the enclosure for a few minutes each evening, no touching (page 18)
- ☐ Confirm it uses the warm hide. If it never does, re-read the floor and check the hide is snug

Weeks 2 to 3

- ☐ Once it stops fleeing and eats normally, 5-minute handling sessions every other day
- ☐ After the first shed, check toes, tail tip and eyes (page 31)
- ☐ Start the Owner Log with a weekly weigh-in (page 41)

Week 4

- ☐ Review the weight and the tail: steady gain for a young gecko, stable weight for an adult (page 17)
- ☐ Settle the feeding interval on body condition rather than the calendar (page 12)
- ☐ Set a phone reminder for the UVB replacement date and a monthly thermostat check
- ☐ Do the first monthly deep clean (page 11)
- ☐ Introduce the first enrichment items (page 43)

Days 31 to 90

- ☐ Book any follow-up the vet recommended, and act on the fecal result rather than filing it
- ☐ Keep any quarantine running the full 3 to 6 months before the gecko shares a room, tools or hands with another reptile
- ☐ Look back at the Owner Log for any slow drift in weight, appetite or tail thickness
- ☐ Fill in the pet-sitter sheet before you need it (page 40)
- ☐ Check the sex again as it matures, and add a lay box for a female (page 22)

Symptom quick reference

SYMPTOM	LIKELY CAUSE	ACTION
Tail thinning to a stick, weight falling	Cryptosporidiosis	Keep apart from other reptiles, vet, fecal PCR (page 27)
Soft or crooked jaw, tremors, bent limbs	Metabolic bone disease	Vet promptly; remove climbing decor (page 28)
Bloated belly, no normal droppings, straining	Impaction	Check temperatures, vet promptly (page 29)
Wrinkled skin, sunken eyes, yellow or gritty urates	Dehydration	Water access and a damp humid hide; vet if it persists (page 29)
Pink or red tissue from the vent	Prolapse	Keep moist, vet immediately (page 29)
Open-mouth breathing, bubbly saliva, clogged nostrils	Respiratory infection	Vet; warm side up, moisture source found (page 30)
Repeated watery stool	Internal parasites	Vet with a fresh sealed sample (page 30)
Tiny dark or reddish flecks, long soaks in the water dish	Mites	White-paper check, vet (page 30)
Shed stuck on toes, tail tip or around the eyes	Low humidity, low vitamin A	30-minute soak and a damp swab; eyes to the vet (page 31)
Tail dropped	Grabbed, frightened, or attacked	Paper towel, keep it clean, no handling for 1 to 2 weeks (page 31)
Blistered, blackened or peeling skin	Burn from unregulated heat	Heat off, fix the thermostat, vet (page 31)
Undigested food, foul-smelling stool	Cryptosporidiosis, or a warm side too cool	Read the floor today, then vet (page 32)
Food refused, gecko bright, tail full	Cool warm side, a shed, brumation	Read the warm hide floor first (page 15)
Quiet, hiding, eating less as days shorten	Possible brumation	Weigh every couple of weeks (page 24)
Slow tail swish, arched back	A warning: leave it alone	Close the enclosure, try later (page 19)
Pacing away from the warm side, gaping, flattening in the coolest spot	Too hot	Read the floor, fix the heat (page 7)
Female restless, digging, belly bulging	Gravid	Lay box now (page 22)
Female straining or weak without laying	Egg binding	Vet the same day (page 23)
Weight loss between log entries	Several possible causes	Vet, bring the Owner Log (page 17)

Daily, weekly & seasonal routine

Daily

- ☐ Read the warm-end thermometer against the thermostat display; a reading that disagrees sharply or keeps climbing is the warning
- ☐ Fresh water, and check the humid hide is still damp
- ☐ Feed in the evening on the schedule for its age (page 12)
- ☐ Dust supplements on the right days (page 16)
- ☐ Remove uneaten insects
- ☐ Spot-clean droppings, and look at them before you do (page 32)
- ☐ Quick look: tail thickness, eyes, toes, posture
- ☐ Wash your hands after every contact

Weekly

- ☐ Weigh in grams and log it (page 41)
- ☐ Read the warm hide floor with the infrared thermometer, and the hygrometer
- ☐ Wipe down glass, walls, hides and decor
- ☐ Replace the humid hide lining if it has started to break down

Monthly

- ☐ Deep clean: everything out, scrubbed, disinfected, rinsed, dried, fresh substrate (page 11)
- ☐ Confirm the thermostat still cuts in, the probe hasn't moved, and the timer is on schedule
- ☐ Review the Owner Log for slow trends, not just single readings
- ☐ Move or swap one furnishing, not the whole layout

Seasonal

- ☐ Fall: expect appetite to drop; weigh at least every couple of weeks through any brumation (page 24)
- ☐ Winter: check the room's night low, and add a ceramic heat emitter on a thermostat if it falls below 65°F (18°C)
- ☐ Adult females: a lay box ready whenever she looks gravid (page 22)

Yearly

- ☐ Wellness exam and a fecal test, even for a gecko that seems perfectly healthy (page 26)
- ☐ Replace the UVB tube by its rated life, 12 months at the latest
- ☐ Replace opened powders by the date on the jar

IF YOU ONLY KEEP ONE HABIT

Keep the weekly weigh-in. Most of what goes wrong in a leopard gecko shows up as a weight change or a thinning tail before anything else, and in an animal this small a slide of a few grams is invisible by eye and obvious on a scale.

Power outages, heat, travel & transport

Normal night low about 70°F (21°C); act below 65°F (18°C). A small gecko loses heat faster than a heavy-bodied snake.

When the power goes out

Do

- Keep the gecko at or above 65°F (18°C). A battery or wind-up thermometer in the enclosure gives you the real number.
- Drape thick towels or a blanket over the enclosure, leaving a vent gap, to hold in what heat is left.
- Move the enclosure, or the gecko in a ventilated tub, to the warmest room.
- Put a wrapped hot-water bottle or a hand warmer beside the gecko, never touching it.
- Stop feeding insects. Offer water only until heat is reliably back.
- Past a day or two below 65°F (18°C), board it with a vet, a friend with power, or a reptile shop.

Don't

- Don't use candles, gas stoves, or open flame near the enclosure.
- Don't put a heat pack in contact with the gecko. A cold, sluggish animal can't move away fast enough and will burn.
- Don't feed "to keep its strength up." Undigested food in a cold gut is a bacterial and impaction risk.
- Don't warm a cold gecko in hot water; lukewarm only, and only once it's alert.

OUTAGE KIT

Two or three hand warmers or a hot-water bottle, two thick towels, a battery thermometer, a ventilated tub, and your vet numbers written on paper.

In the heat

The cool side has to stay at 70 to 77°F (21 to 25°C), and a thermostat cannot cool the room. Gaping, pacing away from the warm side or flattening into the coolest spot means too hot; darkened skin or limpness is an emergency.

Leaving town

TRIP LENGTH	ADULT	JUVENILE
1 to 2 nights	Often fine alone. Lights on timers, thermostat running, fresh water and a damp humid hide before you go.	A sitter once a day for food, water and a temperature check.
3 to 7 nights	A sitter every 1 to 2 days for water, a stool check and a temperature check. Feed on the normal schedule.	A sitter daily, with pre-portioned, pre-dusted insects and written instructions.
Over a week	A daily sitter who has done a practice visit with you, or boarding with a reptile-experienced vet or shop. Never a general dog-and-cat boarding facility. A gecko that is sick, mid-shed, gravid or under a year old gets the juvenile column.	

Transport and vet trips

- A ventilated plastic tub with a towel to grip, in a dark, room-temperature bag or box. No loose substrate, no water dish.
- Warm the car first in cold weather; for a trip over about 30 minutes below 65°F (18°C), a wrapped hand warmer beside the tub. Check the tub every 20 to 30 minutes at a stop.
- Skip the meal before a trip. Never leave the gecko in a parked car, hot or cold, even briefly.
- Moving house: the enclosure runs at the new place before the gecko arrives.

FILL IN AND LEAVE BY THE ENCLOSURE

Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person before the first solo visit.

Gecko's name _____

Age and normal weight _____

My phone _____

Reptile vet _____

24-hour emergency clinic _____

Backup contact _____

Each visit

- ☐ Check the thermometers. The warm hide floor should read 88 to 92°F (31 to 33°C), the cool side 70 to 77°F (21 to 25°C). Lights should be on between _____ and _____.
- ☐ Rinse and refill the shallow water dish.
- ☐ Feel the lining of the humid hide (the closed box on the cool or middle side), and dampen it if it has dried out.
- ☐ Insects on these days: _____. Give about _____ from the tub, by tongs, in the evening. Remove any left after 15 minutes.
- ☐ Dust with the calcium jar on _____ (a light coating, right before feeding).
- ☐ Scoop any droppings with a paper towel. Normal is a firm dropping with a white end.
- ☐ Wash your hands before and after. Don't rinse anything in the kitchen sink.

Please don't

- Take the gecko out of the enclosure, unless we agreed on it, and never pick it up by the tail. It will drop it.
- Change any dial, timer or thermostat setting, or move the lights.
- Feed anything not on this sheet, and never fruit, vegetables or anything caught outside.
- Spray the enclosure with water; only the humid hide is meant to be damp.

CALL ME IF

The lights are off during the day, or the warm hide floor reads well below 88°F (31°C) · the gecko is breathing with its mouth open, or there are bubbles at the nose or mouth · it hasn't moved across two visits, or is limp and hard to rouse · there's runny or bloody poop · anything is protruding from the tail end · the tail has come off · anything else that looks wrong. If I don't answer, call the vet number above.

One row a week. Photocopy this page, or track digitally. Consistency matters more than the format.

WHY BOTHER LOGGING

A steady weight trend is the best early warning you have, and in this species doubly so: a gecko that is brumating holds its weight and its tail, and a gecko that is ill does not, and the log shows the difference weeks before the gecko does. Sixteen rows is four months, long enough to see a real trend.

Equipment & vet log

Equipment log

ITEM	INSTALLED	REPLACE OR CHECK BY	DONE
UVB tube (6 to 12 months, by the model)			
Heat mat and thermostat (check monthly)			
Thermostat probe position (every spot clean)			
Humid hide lining (replace as it breaks down)			
Thermometer and hygrometer batteries			
Calcium and multivitamin opened (by the jar's date)			

Twelve-month planner

Tick the box in the month you did it. Month 1 is whichever month you start in.

TASK, BY MONTH	1	2	3	4	5	6	7	8	9	10	11	12
Deep clean	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Thermostat check	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
UVB tube replaced						☐						☐
Vet exam and fecal	☐											☐

Vet visit log

DATE	WEIGHT (G)	REASON AND FINDINGS	FOLLOW-UP

Enrichment checklist & log

Setup

- ☐ All three hides in place, snug, and the humid one genuinely damp
- ☐ Cork flats, slate or a low sturdy branch to rest on and rub against
- ☐ A dig box, especially for a female that digs
- ☐ Cover that still lets the gecko move through the space

Monthly review

- ☐ Move or swap one item, not the whole layout
- ☐ Vary the feeding: tongs, a short supervised hunt, a rolling dispenser
- ☐ Try a different staple insect (page 13)

Enrichment log

DATE	ACTIVITY	HOW IT WENT, AND WHETHER TO REPEAT

ENRICHMENT SAFETY

Never leave feeder insects loose in the enclosure long-term, never use wild-caught insects, and never treat a shed as handling time. Nothing on this page substitutes for correct belly heat and three working hides, which is where the research on this species keeps pointing.

Glossary, A to H

Autotomy	Caudal autotomy: a lizard dropping its own tail when grabbed or frightened. A leopard gecko's tail regrows in about 30 days, shorter and thicker. Pages 18, 31.
Bioactive	A planted enclosure with a living substrate and a cleanup crew of tiny invertebrates that break down waste. An option for experienced keepers. Page 10.
Brumation	A reptile's natural winter slowdown: less activity and appetite, cued by cooler temperatures and shorter days, without weight loss. Page 24.
Captive-bred	Hatched from parents kept by people rather than caught in the wild. Essentially every leopard gecko sold today is captive-bred. Pages 5, 25.
Ceramic heat emitter	A screw-in heater that gives off heat with no visible light, so it can run at night on its own thermostat. Page 8.
CITES	The Convention on International Trade in Endangered Species of Wild Fauna and Flora: the treaty that controls international trade in listed animals and plants. The leopard gecko is not on its most restricted list, Appendix I. Page 25.
Coccidia	Single-celled gut parasites, found with a fecal test, often with no outward sign. Page 30.
Crepuscular	Most active at dusk and dawn, which is why evening is the time to feed and handle. Page 5.
Cryptosporidiosis	A contagious parasite infection with no cure, called stick tail disease for the wasting it causes. Diagnosed with a fecal PCR test. Page 27.
D3, vitamin D3	The form of vitamin D a reptile makes in its skin under UVB, or takes in from a dust, and needs to absorb calcium. Pages 9, 16.
Deep heat projector	An overhead heater with no visible light that sends its heat down in a more focused beam than a ceramic heat emitter. Runs on a dimming thermostat. Page 8.
Dimming thermostat	A thermostat that lowers the power to a heater the way a light dimmer does, instead of switching it fully off and on. Also called proportional. Page 8.
Dysecdysis	Incomplete or "stuck" shedding, most often on the toes, the tail tip and around the eyes. Page 31.
Dystocia	Egg binding: a female unable to pass her eggs. A same-day emergency. Page 23.
Ecdysis	The normal shed cycle, from dull, dry pre-shed skin to the old skin coming off. Page 24.
Fecal test	A vet's check of a stool sample for parasites, under the microscope or by PCR. Bring a fresh sample in a sealed bag. Pages 21, 30.
Ferguson Zone	One of four groups that match a species' natural sun exposure to a UVB strength. Leopard geckos are Zone 1, the crepuscular and shade-dwelling group. Page 9.
Gravid	Carrying eggs. A female can be gravid without ever meeting a male. Page 22.
Gut-loading	Feeding feeder insects a calcium-rich diet for 24 to 72 hours before offering them, so the calcium in their gut passes to the gecko. Page 14.
Hemipenal bulges	The two bumps at the base of a male's tail where the hemipenes, its paired reproductive organs, sit. Page 22.
Humid hide	A closed hide with a damp lining, 70 to 80% humidity inside, in an enclosure otherwise kept at 30 to 40%. It prevents most shedding problems. Page 10.

REFERENCE

Glossary, I to Z

Impaction	A blockage in the digestive tract, usually from swallowed loose substrate or prey too large to pass. Page 29.
Infrared thermometer	A handheld gun that reads the temperature of a surface from a distance. The way to check the warm hide floor. Page 7.
Lay box	A covered container of damp sphagnum moss, big enough to turn around in, where a female digs and lays. Page 22.
MBD	Metabolic bone disease: a weakened, deformed skeleton from too little calcium, vitamin D3 or heat. Common in leopard geckos and largely preventable. Page 28.
Morph	A captive-bred line that looks different from the natural, wild-type gecko in color or pattern. Over 100 are recognized. Page 5.
Obligate insectivore	An animal that must eat insects and cannot live on plant food. A leopard gecko is one. Page 12.
On/off thermostat	A thermostat that cuts power fully at the set temperature and restores it once the reading drops. The usual match for a heat mat. Page 8.
Oxalates	Compounds, oxalic acid among them, in spinach, beet greens and Swiss chard that bind calcium so it can't be absorbed. Why those greens make a poor gut-load. Page 14.
PCR	Polymerase chain reaction: a lab test that finds a germ's genetic material in a sample. How a vet tests stool for cryptosporidiosis. Pages 21, 27.
Preanal pores	A V-shaped row of waxy pores just in front of the vent, enlarged in a maturing male. Page 22.
Probe	The thermostat's sensor. On the warm hide floor, held with a clip or suction cup, never tape. Page 8.
Prolapse	Tissue pushed out of the vent. Keep it moist and see a vet immediately. Page 29.
Pulse proportional thermostat	A thermostat that sends short pulses of power to hold a steady temperature. Suits a ceramic heat emitter. Page 8.
PVC	Polyvinyl chloride: a rigid plastic used for enclosure panels. Holds heat that a glass tank loses. Page 6.
Quarantine	Keeping a new reptile fully apart from others, in a separate room with separate tools, for 3 to 6 months. Page 21.
<i>Salmonella</i>	Bacteria that healthy reptiles carry in the gut and shed in droppings without being ill. Spread from hands and surfaces to mouths. Page 11.
Stick tail	The common name for cryptosporidiosis, after the tail that wastes down to a stick. Page 27.
T5 HO	A fluorescent tube 5/8 inch across (T5), in the high-output (HO) version. The standard UVB source; coil bulbs are not equivalent. Page 9.
Thermal gradient	The warm-to-cool range across the enclosure that lets a gecko choose its own temperature. Page 7.
Units	°F, degrees Fahrenheit, and °C, degrees Celsius, for temperature; in (inches) and cm (centimeters) for size; g (grams) for weight; % (percent) for humidity.
Urate	The white, chalky part of reptile waste, made from uric acid instead of liquid urine. Its color and texture are a hydration gauge. Page 32.
UVB	Ultraviolet B: the part of sunlight a reptile's skin uses to make vitamin D3. Optional for a leopard gecko, and beneficial at low output. Page 9.
UVI	Ultraviolet index: a measure of UVB strength at one spot, read with a handheld meter. 0.6 to 1.4 at the basking spot for this species. Page 9.
VCA	Veterinary Centers of America, now VCA Animal Hospitals: an animal hospital chain whose pet care articles are among this package's sources.

REFERENCE

Sources

Where the numbers in this package come from, so you can check them yourself.

Veterinary references

- PetMD: Leopard Gecko Care Sheet, and Stick Tail Disease in Leopard Geckos. LafeberVet: Basic Information Sheet, Leopard Gecko. Adult size and weights, lifespan, temperatures, and the cryptosporidiosis outlook.
- VCA Animal Hospitals: Leopard Geckos, Care and Feeding; Reptiles, Dystocia; the new reptile visit; choosing a pet reptile. MedVet: Leopard Gecko Information and Care Sheet. Safarivet: Feeding Leopard Geckos. Yarmouth Veterinary Center: The Gravid Gecko.
- Merck Veterinary Manual: management and husbandry of reptiles; disorders and diseases, parasitic, viral and bacterial diseases of reptiles; clinical procedures; selecting a reptile; cleaning and disinfection. Thermostat control, the 3 to 6 month quarantine, and the bleach dilution.
- Kevin Wright, veterinarian, in Reptiles Magazine: diagnosing and treating retained shed. The 30-minute soak and the six causes of a bad shed. DoveLewis Emergency Animal Hospital: straining as a warning sign.
- Association of Reptile and Amphibian Veterinarians: the find-a-vet directory.

Research

- Gould and colleagues (2018), Journal of Herpetological Medicine and Surgery 28(1-2): blood vitamin D in leopard geckos given ultraviolet light or dietary vitamin D. Oonincx and colleagues, Wageningen University: vitamin D3 made in the skin under UVB.
- Nipithakul and colleagues (2025), Animals: *Cryptosporidium varanii* infection in captive leopard geckos and wasting in Thailand.
- Bashaw and colleagues (2016), Applied Animal Behaviour Science 184: five types of enrichment. Rickman and colleagues (2025), University of Lincoln: enriched housing and welfare.
- Baines and colleagues (2016), Journal of Zoo and Aquarium Research 4(1): How much UV-B does my reptile need? The Ferguson Zone framework. Rhen and colleagues (2011), Heredity: temperature-dependent sex in leopard geckos.
- Finke (2002 and 2013), Zoo Biology: the nutrient content of feeder insects. Allen and Oftedal (1989) and Klasing and colleagues (2000), Journal of Zoo and Wildlife Medicine: raising feeder calcium by diet.

Public health

- Centers for Disease Control and Prevention: Healthy Pets, Healthy People, reptiles and amphibians, and the *Salmonella* Muenchen outbreak linked to pet geckos, closed January 2026. Center for Food Security and Public Health, Iowa State University: reptile- and amphibian-associated salmonellosis, and isolation and quarantine areas.

Husbandry, equipment, costs and the law

- ReptiFiles: Leopard Gecko Terrarium Size (the 36×18×18 in minimum); Leopard Gecko Handling and Body Language; Leopard Gecko Respiratory Infection. Zen Habitats: leopard gecko lighting and heating, overhead versus under-tank heat, signs of overheating, and the leopard gecko food guide. Spyder Robotics, thermostat manual: probe placement and the warning against tape.
- The Vet Desk and Reptile Direct: leopard gecko cost breakdowns. Retail prices checked in September 2026.
- Hawaii Administrative Rules § 4-71-6.5; District of Columbia Code § 8-1808(j); New Jersey Administrative Code 7:25-4.3 and 4.6; Maine Department of Inland Fisheries and Wildlife, Unrestricted Species List; Minnesota Statutes § 97A.401; West Virginia Code § 20-2-13; and each other state's rules, read on 1 October 2026 (Maine on 5 September 2026).
- Beastly Facts guides, current at publication: the leopard gecko enclosure setup, temperature, feeding, handling, health issues, enrichment, cost and legal guides, the UVB lighting and T5 articles, and the cross-species reptile quarantine, heating and thermostats, emergency plan, stool and urates, gut-loading, *Salmonella* and hygiene, and shedding guides.

REFERENCE

Where the sources disagree

Where credible sources give different numbers, this package prints one and says why.

TOPIC	WHAT THE SOURCES SAY	WHAT THIS PACKAGE USES
Adult length	6 to 9 in (a veterinary care sheet); 6.5 to 8 in (a veterinary information sheet)	6 to 9 in (15 to 23 cm)
Lifespan	10 to 20 years typical; some individuals reach 25 to 30	10 to 20 years, some reaching 25 to 30
Adult feeding interval	Every other day; every 2 to 4 days; every 5 days	Every 2 to 4 days, set by the tail and the weight
An adult off food	A couple of weeks is fine with a plump tail; a month at the outer edge; a vet if appetite loss persists more than a few days	Watch the tail, not the calendar; vet at once with a thinning tail or any other sign
Gut-loading	24 to 48 hours; 24 to 72, with 48 most cited	24 to 72 hours, 48 typical
Temperature gradient	70 to 80°F with a warm area reaching 90°F, up to a 90 to 92°F warm hide over a 70 to 77°F cool side	Warm hide floor 88 to 92°F, air 80 to 84°F, cool side 70 to 77°F
Night floor	From 60°F up to the low 70s	Nights in the 70s, 65°F the floor
Overhead or belly heat	Belly heat preferred; overhead heat also works and many keepers run both	A mat under the warm hide, overhead optional
Quarantine	3 to 6 months (veterinary); 60 to 90 days (a hobby floor)	3 to 6 months
UVB tube life	6 to 12 months by model; roughly once a year for most quality T5 tubes	The maker's rated life, 12 months at the latest
Stuck-shed soak	Shallow lukewarm water; chin-deep at enclosure temperature for 30 minutes (a reptile veterinarian)	Shallow, at enclosure temperature, chin-deep at most, 30 minutes
Damp lining	Humid hide: damp substrate, not sphagnum moss or paper towel the gecko could swallow; lay box: damp sphagnum moss	Each as its own guide says
How common is stick tail	"About half of captive geckos" (an often-repeated estimate); 51% of geckos at one Thai breeding facility (a 2025 study)	The single study, named as one facility

REFERENCE

Version history & about

What changed in each edition, so a printed copy can be placed.

Version history

This is edition 3.0 (Oct 2026). The store page shows the current edition; re-download free if yours is older. Where this list and a page disagree, the page wins.

- **3.0, Oct 2026.** Rebuilt on the current outline, 48 pages. Safety: thermostat probe held with a clip or suction cup, never tape; lamps at least 4 to 6 in above the gecko; cool side 70 to 77°F (21 to 25°C); nights in the 70s with 65°F (18°C) the floor, and the outage floor raised from 60 to 65°F (18°C); UVI 0.6 to 1.4 at the basking spot; UVB tube replaced every 6 to 12 months by model; first vet exam within 72 hours; a slow tail swish with an arched back is the warning sign, and a rapid tail flick is excitement, not distress; stuck shed loosened with a 30-minute soak; adult length 6 to 9 in; lifespan 10 to 20 years, some reaching 25 to 30. Legal: a new page on the law in all 52 jurisdictions. Costs: monthly running costs now \$20 to \$50 for most owners, with the yearly exam as its own line. Other changes are wording only.
- **2.1, Sep 2026.** Safety: quarantine for a new gecko raised to 3 to 6 months in a separate room; a second thermometer read daily; full *Salmonella* hygiene rules; dehydration soak 15 to 20 minutes. Other changes are wording only.
- **2.0, Sep 2026.** Rebuilt, 34 pages. Safety: calcium with D3 as a backup for setups without UVB, not a routine dust.
- **1.0, 2026.** First edition, 22 pages. Replaced by 2.0.

This package is a husbandry reference compiled from the listed sources and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for a reptile-experienced veterinarian, and nothing in it should delay a vet visit. Treatments described here are for recognizing what a vet may do, never for use at home. The law page is a summary as of October 2026, not legal advice.

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